

Experient 1 Private Cloud using Openstack

Aim:

To design, install, and configure a Private Cloud environment using a single-node OpenStack (MicroStack) implementation, and to understand the basic cloud services such as compute, networking, and storage by deploying and accessing virtual machine instances on the private cloud.

Openstack:

What is openstack?

1. OpenStack is a free, open standard cloud computing platform.
2. It is mostly deployed as infrastructure-as-a-service(IaaS)
3. OpenStack allows to deploy virtual machines (VMs) on demand.

It controls large pools of

1. compute 2) storage 3) networking resources of a datacentre

-managed by admins via dashboard -provides resources to users via web interface.

Private/Hybrid cloud->OpenStack. Public cloud ->AWS,Azure...

what is microstack?

MicroStack runs OpenStack locally with minimal setup(single-node).

Ideal for learning, testing, or small development projects.

It's easy to deploy and test OpenStack on a single machine. (like a laptop or small server).

openstack services:

1. Nova (Compute)
2. Neutron (Networking)
3. Keystone (Identity)
4. Glance (Images)
5. Horizon (Dashboard)

Function	OpenStack	AWS	Description
Compute (VMs)	Nova	EC2	Run virtual machines
Image Management	Glance	AMI	Store and manage VM images
Identity & Access	Keystone	IAM	User authentication, authorization, roles
Networking	Neutron	VPC, SecurityGroups	Manage networks, routers, IPs, firewalls, load balancing
Block Storage	Cinder	EBS	Persistent volumes for VMs
Object Storage	Swift	S3	Store and retrieve large amounts of unstructured data
Dashboard (UI)	Horizon	AWS Console	Web interface for managing cloud resources

[OPENSTACK\(microstack\) installation steps in Ubuntu 22.04 LTS\(jammy\)](#)

sudo snap install microstack --beta

snap list microstack

sudo apt install python-is-python3

sudo microstack init --auto --control

type→open browser and type 10.20.20.1 dashboard appears

username=admin

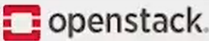
//for password

sudo snap get microstack config.credentials.keystone-password

10.20.20.1/auth/login/?next=/

openstack.
Log in
User Name

Password

10.20.20.1/project/
 openstack. admin
Project
API Access
Compute
Overview
Instances
Images
Key Pairs
Server Groups
Volumes
Network
Admin
Identity
Project / Compute / Overview
Overview
Limit Summary
Compute
Instances
Used 0 of 10
VCPUs
Used 0 of 20
RAM
Used 0Bytes of 50GB
Volume
Volumes
Volume Snapshots
Volume Storage

Result

A single-node Private Cloud environment was successfully set up using OpenStack (MicroStack). The OpenStack services were properly installed and configured, and the private cloud was able to launch, manage, and access virtual machine instances with functional compute, networking, and storage services, thereby meeting the objectives of the experiment.